MIL-C-14027A(WC)
30 October 1964
SUPERSEDING
MIL-H-14027(Ord)
6 APRIL 1955

MILITARY SPECIFICATION

CANNON, 155MM HOWITZER: M45

1. SCOPE

1.1 This specification covers one type of howitzer cannon.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-11472 - Magnetic Particle Inspection, Process for Ferromagnetic Material

MIL-C-13931 - Cannons, Mortars and Recoilless Rifles and Guns, General Specification for

DRAWINGS

U. S. Army Weapons Command

EPL 7309247 - Engineering Parts List for Cannon, 155MM Howitzer: M45

PUBLICATIONS

U. S. Army Weapons Command

LSQAP 7309247 - List of Supplementary Quality Assurance Provisions for Cannon, 155MM Howitzer: M45

LPDS 7309247 - List of Packaging Data Sheets for Cannon, 155MM Howitzer: M45

FSC 1025

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and construction. Cannons and parts thereof shall conform to this specification, the drawings and specifications listed in Engineering Parts List EPL 7309247 and Specification MIL-C-13931.

3.2 Breech mechanism. All parts of the breech mechanism shall operate without interference, erratic movement or malfunction. With the breech in the closed position a rearward and downward pull on the operating lever, after release of the latch, shall rotate the breechblock clockwise to disengage the sectored threads. A pull to the right on the operating lever shall cause the breechblock to swing out of the breech to the open position. The breechblock shall be retained in the open position by the counterbalance. To close the breech, a push on the operating lever to the left, past the dead center position of the hinge pin arm, shall swing the breechblock into the breech opening. This closing action shall cause the roller on the breechblock to enter the cam on the breech ring thereby rotating the breechblock counterclockwise to engage the sectored threads. An upward push on the operating lever shall continue the rotation of the breechblock and lock the breech in the firing position.

3.3 Proof testing

3.3.1 First of production. The first cannon and repair tube manufactured under a contract shall be subjected to the production engineering tests specified by the responsible commodity Arsenal and the firing tests specified in 4.3.

3.3.2 Regular production. The cannons and repair tubes after the first of production shall be capable of withstanding the proof firing tests specified in 4.3.

3.3.3 Material soundness. After the firing tests, all parts shall be free of cracks, fractures and tears as evidenced by visual examination and magnetic particle or liquid penetrant testing.

3.3.4 Bore enlargement. Bore diameters, measured across rifling lands after proof firing no more than 9 rounds, shall not exceed the basic (minimum) diameter specified on the applicable drawing by more than the amount specified in Table I.

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TABLE I

Distance forward of commencement of rifling	0.1 Inch	1-Inch	1-Cal.	2-Cal. to Muzzle
Amount (Inch)	0.007	0.007	0.005	0.004

NOTE: Interpolate linearly to determine allowable bore enlargements at distances other than the distances listed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection performance. Inspection shall be performed in accordance with MIL-C-13931 except as otherwise modified or specified herein.

4.2.1 AQL values. The acceptable quality levels (AQL's) are set forth in the SQAP's listed in LSQAP 7309247.

4.2.2 Functioning of breech mechanism. Manually operate all parts of the breech mechanism of each assembled cannon a minimum of two complete cycles of operation. Any evidence of binding, scraping, looseness or other malfunction of any part shall be cause for rejection.

4.3 Proving ground testing. Cannons and repair tubes supplied under this specification shall be subjected to proving ground inspection in accordance with the applicable inspection plan at a test agency designated by USATECOM. Proof firing and inspection shall be performed by Government personnel.

4.3.1 Proof firing procedures. The first cannon and the first repair tube manufactured under a contract shall be proof fired in accordance with Table II. Subsequent cannons and repair tubes shall be fired in accordance with Table III.

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TABLE II

Number of Rounds	Per Cent of Rated Maximum Pressure
1	50
1	75
1	100
2	115
1	100
2	115
1	100

TABLE III

Number of Rounds	Per Cent of Rated Maximum Pressure
1	100
1	115

4.3.2 Proof firing rate. Proof firing shall be conducted at a rate of fire not to exceed one round per minute.

4.3.3 Exterior surfaces. After proof firing, examine the outer surfaces of the tube, breech ring, breechblock and carrier in accordance with MIL-M-11472 to determine compliance with the soundness requirements specified in 3.3.3.

4.3.4 Proof marking. Proof acceptance marking shall be located on the tube near the serial number.

4.4 Inspection of preparation for delivery.

4.4.1 Examination. The classification of defects and the AQL's for examination of the packaging, packing and marking are as follows:

Categories	Defect	AQL
Critical:	None	
Major		
101	Illegible or incorrect marking	1.0
102	Improper level of packaging and packing (see procurement documents)	1.0
103	Inadequate cleaning and drying	1.5
104	Improper preservative application	1.5

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Categories	Defect	AQL
Major (Continued)		
105	Missing or improper barrier material	1.5
106	Improper or inadequate blocking and bracing	1.5
107	Improper closure and strapping of container	1.5
Minor:	Workmanship of container	4.0

5. PREPARATION FOR DELIVERY

5.1 Cannons and repair parts. Cannons and repair parts shall be prepared for delivery in accordance with the packaging data sheets listed on LPDS 7309247 for the level of protection specified in the procurement documents.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Serial numbers for the cannons and the repair tubes.

(c) List of repair parts required.

(d) Production engineering tests on first of production cannon or repair tube.

(e) Availability of inspection equipment from ordnance stocks.

(f) Extent of suppliers' responsibility for Government-furnished and supplier-acquired final inspection equipment.

(g) Levels of packaging and packing.

NOTE: Level "B" is intended to provide adequate and economical protection for multiple domestic shipments, handling, and indoor storage periods up to six months from date of initial packaging.

(h) When a pilot pack is required.

(i) That packages opened for acceptance inspection shall be repackaged at the supplier's expense.

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6.2 Storage surveillance. Cannons and repair tubes packaged and packed in accordance with Level "B" requirements must be inspected to determine condition when not used within a six-month period. Cannons and repair tubes not used within this period shall be represerved, repackaged and repacked in accordance with this specification, as determined by storage conditions and anticipated storage time.

6.3 Certain provisions of this specification are the subject of international standardization agreement ABC-ARMY-STD 98. When amendment, revision or cancellation of this specification is proposed, the Departmental Custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodian:

Army--WC

Preparing activity:

Army--WC

Project No. 1025-A056

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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